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Palaeomagnetic Studies on Basic Dykes and Sills from S.W. England

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Summary

Numerous diabase intrusions have been mapped in L. Carboniferous and Devonian Sediments of S.W. England. These have been described as sills or dykes, depending on whether they are supposed to have been intruded before or after folding. However it is not always possible to decide this for certain from usual geological criteria. Also there has been some discussion as to when these igneous bodies were in fact formed. Palaeomagnetic studies, described in this paper suggest that these diabase intrusions were remagnetized in their present positions in the U. Carboniferous or Permian. Hence they do not help resolve the geological problem as was originally hoped.

1. Geology

The overall geological structure of Cornwall and Devon was impressed by the Armorican Earth movements. Indurated Devonian and Carboniferous sediments, deposited in a post Caledonian geosyncline were folded and compressed to form a huge compound geosyncline with a general east–west orientation. The Carboniferous Culm Measures lie at the centre. They are isoclinally folded and are flanked on the north and south by Devonian rocks which are sometimes overturned.

After the folding a series of granite domes were intruded along the same east–west axis. These intrusions are complex, each consisting of a number of large sheet like masses of slightly different composition. They have been dated by the potassium–argon method (Miller & Mohr 1964). Fourteen determinations yielded an average age of 282 ± 11 m.y.

Both Devonian and Carboniferous sediments are interstratified by a large number of basic dykes and sills. It is not known whether they were intruded before or after folding, different geologists who have worked in different, but not distant localities, having arrived at different conclusions. They are, however, pre-granite, since when they are examined in outcrops close to the granite, they appear to have been metamorphosed and contain abundant pyrrhotite. Potassium–argon ages vary from 372 ± 22 m.y. (for an albite diabase from Newlyn, Land's End) through 337 ± 27 m.y. for a dolerite near Lower Ashton, S. Devon to 296 ± 12 and 295 ± 6 m.y. for type I and type II dolerites at Meldon. This range of ages is about the same as that of the sediments into which they are intruded.

Upper Devonian rocks are intruded also by large amounts of spilitic pillow

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lavas in which the ferromagnesian minerals are almost completely converted to chlorite.

2. Purpose of present work

The original purpose of these palaeomagnetic studies, started in 1958, was to attempt to resolve the problem of whether the basic dykes and sills were intruded before, during or after folding. This is not clear in this region from usual geological criteria. The question often arises of whether the rocks are the right way up or upside down, both in studies of the sediments and igneous rocks. It is not certain, on geological evidence, whether igneous sheets intruded into now almost vertically dipping sediments are sills intruded before folding or dykes intruded after. For instance N. E. Butcher, of Reading University has argued that the former is the case in the Meldon region whereas W. R. Dearman of Newcastle University, working about 10 miles to the south west nearby Tavistock argues the latter case (Dearman & Butcher 1959). Furthermore the emplacement of the granites involved large deflections in the strikes of adjacent rocks, up to 45°. Hence rotations back to the horizontal are very complicated. It was hoped that palaeomagnetic studies might be of use in this kind of problem by the establishment of a direction of magnetization in rocks unaffected by the granites.

The line of investigation planned was thus twofold. First, to measure the directions of magnetization with respect to the present horizontal and then to compare the grouping with that of the same population of vectors corrected for geological tilt. If the sills and dykes were intruded after folding, the former grouping should be the better. If they were intruded before folding the latter grouping of magnetic directions should be the better.

However, the foregoing argument supposes that the primary magnetization, a t.r.m. acquired when these igneous rocks first cooled, was not subsequently destroyed and that the rocks have not been remagnetized. As it turns out, they probably have.

3. Experimental results

A regional survey of the magnetic directions of the basic dykes and sills of S.W. England was made in 1958. This comprises sites 1–18 in Table 1. The positions of these sites are shown in Fig. 1. Some of these sites were briefly resampled in a later survey (Cornwell 1965) and the repeat measurements are indicated by an asterisk. Sites 19–24 were not sampled in the initial survey, but Cornwell's measurements are listed in Table 1. 258 hand samples were collected in the first survey. From six of these eighteen sites, sixteen samples were collected by Cornwell in the later check survey. Cornwell's measurements for the sites 19–24 are based on a total of eighty-one hand samples, and will be described in more detail in a later communication.

The directions of natural remanence and various statistics thereof are listed in Table 1. In Table 2 are listed the range of intensities of natural remanence determined at each site. It will be noted that many of the samples were very weakly magnetized, and had to be measured with the sensitive magnetometer usually used for sedimentary rocks.

The directions with respect to the present horizontal are plotted on an *equatorial* stereographic projection in Fig. 2. Although the scatter within each site is very high, the site-mean directions for the eighteen sites of the original survey comprise

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Table 1

Site No.	Locality	Rock type	Intruded into	D(°)	I(°)	α(°)	K	N	R
1	Crockham quarry Trusham (850809)	Albite Diabase sill	L. Carboniferous sediments	191·4	−38·9	4·0	116·9	12	11·9059
2	Christow quarry (lower) (850850)	Albite Diabase sill	L. Carboniferous	213·1	+70·9	—	5·8	2	1·8262
3	Christow quarry (upper) (840860)	Albite Diabase sill	L. Carboniferous	234·4	−34·4	29·6	3·1	12	8·4750
4	Buckfastleigh quarry (740660)	Greystone	U. Devonian	164·8	−22·2	24·0	5·5	9	7·5573
5	White Cleaves quarry (740680)	Diabase	U. Devonian	234·6	−23·1	41·5	2·5	9	5·7979
6	Torr quarry (745483)	Diabase sill	L. Devonian	190·9	+19·8	30·0	2·1	21	11·4758
7	Hoare Bros quarry (500761)	Diabase sills	L. Carboniferous	207·2	−20·7	16·8	2·8	39	25·6579
8	Brent Tor Tavistock (530810)	Diabase crags	L. Carboniferous	185·2	+14·5	11·8	17·7	10	9·4915
9	Devon C.C. quarry Tavistock (490756)	Diabase sill	L. Carboniferous	194·7	+1·3	11·3	6·4	30	25·4932
10	Cataclews, Padstow (869760)	Basic intrusion Proterobase	U. Devonian	196·8 *185	+3·1 −3·9	50·8 —	1·3 3·0	28 3	6·7760 —
11	Rock, Padstow (929759)	Basic intrusion Minverite	U. Devonian	263·5 *202	−8·5 +7	23·0 —	3·1 4·1	19 3	13·1973 —
12	Penlee quarry Penzance (490305)	Diabase	L. Devonian	197·0	+52·4	43·2	1·8	14	6·8461
13	Polyphant (262820)	Basic intrusion Picrite	U. Devonian	184·0 *204	+27·3 +69	— —	3·2 4·6	3 3	2·3779 —
14	Greystone quarry (380810)	Dolerite intrusion	L. Carboniferous	181·6 *175	+12·2 −2	26·6 37·6	2·9 11·8	16 3	10·8243 —
15	Two Bridges quarry (305817)	Dolerite sill	U. Devonian	213·3 *189	+33·5 −5	27·3 —	2·8 —	16 —	10·6332 —
16	Trevone (889763)	Proterobase	U. Devonian	189·1 *209	−1·2 −5	10·9 —	72·2 7·8	4 2	3·9584

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Table 1—*continued*

Site No.	Locality	Rock type	Intruded into	D(°)	I(°)	α(°)	K	N	R
17	Rumps Point (933813)	Proterobase	U. Devonian	201·2	+9·5	14·3	16·0	8	7·5633
				*202	−10	—	9·7	2	—
18	Dinas Head (865760)	Proterobase	U. Devonian	217·5	+22·2	35·4	4·5	6	4·8946
				—	—	—	—	—	—
19	Clicker Tor (288612)	Ophitic dolerite	U. Devonian	*191	+66	—	4·4	3	—
20	Smeardon Down (518779)	Dolerite sill	L. Carboniferous	*192	−21	31	16·6	4	—
21	Pitts Cleave (501761)	Dolerite sill	L. Carboniferous	*198	−15	14	9·6	13	—
22	Meldon 860S	Dike 1 dolerite	L. Carboniferous	*250	+49	25	15·6	5	—
		Dike 2 dolerite		*288	+35	57	3·6	4	—
		Sediments		*270	+20	32	3·2	10	—
	Meldon 950S	Dike 1 dolerite		*138	+17	34	4·1	7	—
		Dike 2 dolerite		*123	+29	9	37·6	7	—
		Sediments		*156	+41	25	4·0	12	—
	Meldon 1010 (580935)	Sediments		*271	+47	46	2·7	7	—
23	Sticklepath quarry (630940)	Dike 1 dolerite	L. Carboniferous	*210	−25	19	41·0	3	—
24	River Taw (630937)	Dike 2 dolerite	L. Carboniferous	*159	+12	26	7·9	6	—

Measurements which are asterisked comprise later surveys by Cornwell (1965). They are not included in the calculation of a mean direction (Table 3). The agreement of the repeat surveys of sites 10, 11, 13, 14, 15, 16 and 17 suggests that the site mean values are more accurate than the statistics α and K indicate. Site 22 at Meldon is not included in Fig. 2 and will form the subject of a paper to be published soon by Cornwell.

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Table 2

Site No.	Intensities (Gauss × 10^{-6})
1	43·8–73·6
2	
3	1·8– 1·9
4	0·4– 5·2
5	0·5– 2·0
6	1·2– 1·6
7	2·1– 7·3
8	125–140
9	0·3–96·1
10	0·5– 1·4
11	31·6–78·7
12	1·0– 3·3
13	139·5–189·5
14	0·7–23·4
15	4·3–28·1
16	7 –57
17	1 – 3
18	0·5– 1
19	1 –3000
20	2 –10
21	1 –2000
22	
23	26 –78
24	73 –297

Table 3

Population of site means
(*sites* 1–18)

D	I	K	δ	δ_m	N
202°·4	+6°·7	5·01	36°·8	8°·7	18

K, dispersion; δ, circular standard deviation (angular dispersion); δ_m, circular standard error.

Table 4

Site No.	Locality	Rock type	Intruded into	D	I	K
S1	Chipley quarry 807722	Pillow lavas	U. Devonian	232	+46	1·7
S2	Pentewan (Mevagissey) 025477	Spilite	L. Devonian	209	+31	3·4
S3	Two Bridges 271817	Spilite	U. Devonian	220	+42	1·8
S5	Lewannick 267814	Spilite	U. Devonian	162	+13	1·4
M1	Pendennis Point (Falmouth) 827316	Mica lamprophyre	L. Devonian	226	+50	3·7

a well defined group, whose statistics are given in Table 3. The mean direction $D = 202°$, $I = +7°$ for the eighteen sites does not differ appreciably from the lower Permian direction as given, for instance by the Exeter Traps $D = 189°$ $I = -9°$ (Creer 1955, 1957) and the Whin Sill $D = 188°$, $I = -5°$ (Creer, Irving & Nairn 1959). The former have recently been dated at 281 ± 11 m.y. (Dunchideock)

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and 279 ± 6 m.y. (Killerton) (Miller & Mohr 1964) and the latter at about 300 m.y. (Miller & Fitch 1964). This direction is typical of European Permian rocks (Creer *et al.* 1959) and may even have persisted from the upper Carboniferous, because the Whin Sill now appears to be a little older than Lower Permian. Site mean directions for five sites of spiltic lavas and mica lamprophyres (see Table 4) fall into the above group of directions and are indicated by triangles in Fig. 2.

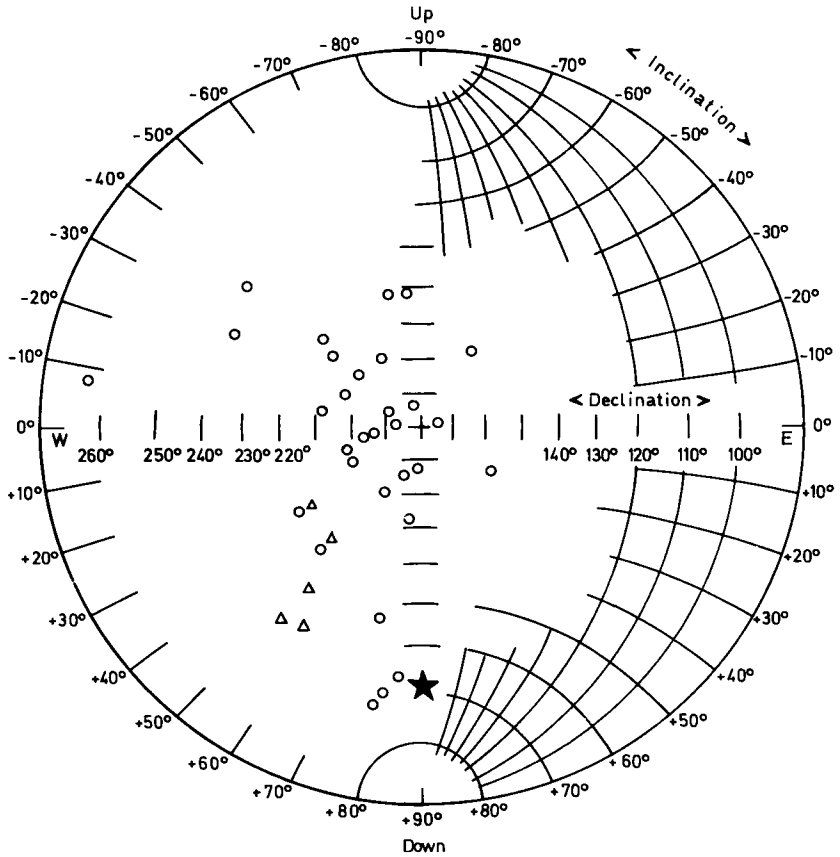


FIG. 2. Site directions of magnetization of diabase (○) and spilites (△). Equatorial stereographic projection. Present dipole field shown by star.

4. Comments on results

The directions of magnetization form a well defined group when measured with respect to present geographic north and the present horizontal (Fig. 2, Table 3). This is consistent with the supposition that the rocks acquired their magnetization while they were in their present positions, i.e. after folding and after the intrusion of the granite.

The directions, corrected to the bedding of the adjacent sediments have not been computed, because it was obvious from the variety of dips and strikes shown on the geological map, that the scatter of such a population of directions would be enormously greater than that shown in Fig. 2. The geological sketch map shown in Fig. 1 indicates this in some measure, although this map has been simplified to such an extent that the geological complexity of the area is not entirely obvious.

The scatter of the site mean values is due to a large extent to a contribution from the within site scatter (Watson & Irving 1957). The dispersion K for the site

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means is 5·0. The log mean dispersion for the within site scatter is 6·5. This mean is taken because the distribution of log K is almost normal whereas the distribution of K values is not. If K_T is the total (measured dispersion), K_{BS} the between site scatter and K_{WS} the within site scatter,

$$\frac{1}{K_T} = \frac{1}{K_{BS}} + \frac{1}{K_{WS}}.$$

Thus the between site scatter is quite small, $K_T = 21{\cdot}7$, in fact it is of the order that one might expect from secular variation (Creer 1962). The circular standard deviation (angular dispersion) given by $\delta = \sqrt{(2/K)}$ radians $= 81/\sqrt{K}$ degrees is 17°·4.

Because the within site scatter is so large, one might ask whether it could not be reduced by a.c. or thermal demagnetization. A.c. demagnetization was carried out on disks from thirty-two hand samples in a pilot experiment. The positive inclination steepens but this change is not significant. There was no reduction in the initially high scatter. This was largely because the already weak rocks became weaker still on demagnetization, and difficult to measure precisely. The relevant data are given in Table 5.

Table 5

A.c. demagnetization of thirty-two samples chosen at random

	D	I	δ	δ_m	N
Before cleaning	221	+3	63°	11°	32
After cleaning	214	+23	73°	13°	32

These data alone are not sufficient to establish the reliability of the mean direction, the main evidence for this as yet being the statistics of Table 3, the large number of samples and sites enabling a reliable mean to be obtained from highly scattered hand sample directions. However the extent of possible remagnetization by the present field is not yet fully assessed.

Further investigation of the stability of the n.r.m. and of the effects of metamorphism by the granites have been made by Cornwell (1965) and these will be published separately, because they reveal a complicated magnetic history at sites in the immediate vicinity of the granites. Evidence taken from Cornwell (1965) given in Table 6, would seem to indicate that the mean direction obtained for the

Table 6

			D	I	α	K	$\bar{J}$	$\bar{K}$	N
21	Pitts Cleave	Dolerite	198	−15	14	9·6	543	88	13
	(501761)	Sediments	181	−28	16	23·1	17	13	5
		Both before cleaning	193	−19	11	10·4	—	—	18
		Both after cleaning	194	−19	9·5	14·3	—	—	18
20	Smeardon Down	Dolerite	192	−21	31	16·6	6	59	4
	518779 & 522779	Calc-flintas	186	−19	—	—	348	178	1
	Wilminstone	Dolerite sills	207	+10	59	2·2	31	48	6
		Sediments	196	−34	18	47	2	26	3
	(490755)	Both before cleaning	204	−18	31	4·1	—	—	8
		Both after cleaning	197	−26	13·2	18·7	—	—	8

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survey is meaningful. Directions from the dolerites and adjacent sediments agree well and there is no significant change in direction after a.c. cleaning.

5. Conclusions

These igneous rocks are much less strongly magnetized than other diabases; the remanent intensity of the Whin Sill, a doleritic intrusion in N. E. England of roughly the same age falls within the range 2 to 4×10^{-3} gauss, as compared with the values listed in Table 2. This anomaly has been the subject of further investigations (Cornwell 1965) to be described soon in this journal.

The mean of site-mean directions of magnetization (Table 3) is close to that of the L. Permian palaeomagnetic field for Europe. The inclination is slightly positive whereas that of the aforementioned direction is slightly negative. This could point to a magnetic age slightly older than the L. Permian.

The important point to note, however, that this mean direction was obtained without making any correction for the dip and strike of the adjacent steeply dipping sediments. This means that the n.r.m. was acquired after folding. N. E. Butcher (personal communication) is fairly certain that at least some of the intrusions in the vicinity of the Dartmoor granite are sills, intruded before folding. Their primary magnetization therefore should have been acquired before folding. Hence the measurements indicate that sills such as these must have been remagnetized after folding took place. The anomalously low intensity suggests that the n.r.m. of these dykes and sills is not an ordinary t.r.m. and it is quite possible that all of them were remagnetized at the end of the Palaeozoic.

This fits in with the suggestion (Chamalaun & Creer 1964) that the O.R.S. of the Anglo-Welsh Cuvette was remagnetized in the L. Permian. The secondary magnetization of this formation completely masks the primary in the n.r.m. It seems likely (Creer 1964) that remagnetization of Palaeozoic rocks, possibly during or after the Armorician Earth movements was widespread in the Laurasian continents.

The intrusion of the granites adds an extra complication to the magnetic history of the rocks of S.W. England. This question has been investigated by Cornwell (1965), particularly in rocks from the Meldon area, on the N.E. boundary of the Dartmoor granite. He finds that when the n.r.m. of the diabase intrusions of S.W. England are studied in detail, the palaeomagnetism of this region is seen to be as complex as the geology. Nevertheless, the main conclusion of this paper remains valid: it is intended to serve as an introduction to later work.

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